Ayantik Sarkar

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Education

National Institute of Technology, Rourkela — Rourkela, Odisha
Bachelor of Technology in Civil Engineering — *2025 – Expected 2029*

Experience

Co-founder & CTO — Dec 2025 – Present
HoardSpace | *hoardspace.in* — *Remote*

- Led a team of 3 engineers building HoardSpace, an outdoor advertising marketplace for 30+ billboard owners managing 3K+ hoardings, which generated Rs. 6+ lakhs in revenue within 3 months of marketing
- Designed per-IP rate limiting (Upstash Redis, in-memory fallback), a honeypot banning scrapers for 24h without blocking Google's crawlers, and prompt-injection defenses that isolate untrusted page context in AI chat
- Architected the core backend: RESTful APIs, Razorpay payments, authentication, and the booking engine
- Set up GitHub CI/CD and built a role-based admin dashboard for inventory and booking management
- Developed HoardSpace AI, a campaign planning assistant (Anthropic models on AWS Bedrock) that recommends ad locations and estimates campaign cost and reach from budget, audience, location, and goals

Projects

Keel | *Next.js, MongoDB, Clerk* — keel.hoardspace.in

- Created an Infisical-inspired secrets manager, used internally at HoardSpace, to replace insecure .env sharing, with secret CRUD, Development/Staging/Production isolation, and version history
- Implemented role-based access control for project permissions, with audit logging of configuration changes
- Developed a CLI runtime-injection workflow (*keel run -- npm run dev*) to run applications locally without exposing raw environment files

Threadpool | *Next.js, MongoDB* — threadpool.in

- Built an agent governance system to register, monitor, and audit AI agents, with an external-agent registry
- Implemented RBAC, policy enforcement, and circuit breakers against runaway agents and excess token use
- Shipped audit trails, API keys, webhooks, and agent workflows; integrated AI models including Sarvam AI

FormAS | *Node.js, Express, Upstash Redis* — formas.space

- Built a form-backend API for developers (no server code, API-key auth), used by HoardSpace's contact form
- Added Redis rate limiting and spam protection, webhooks, email notifications, and JSON/CSV export

Elimics | *React, Node.js, MongoDB* — elimics.vercel.app

- Created a URL shortener with password protection, link expiry, and click-limit controls
- Engineered real-time click, geography, and device analytics with redirect latency under 100ms

Achievements

- **Winner**, Google TechSprint Hackathon 2026 – built EchoCare, which records doctor-patient conversations and uses Gemini to convert them into structured SOAP notes, with patient data encrypted at rest
- **Winner**, Algorand Track, Innofusion 3.0 (Kolkata, 2026; 30+ finalists of 2K+ entries) – built a multi-agent escrow system where a manager agent funds competing agents, pays on completion, and logs every transaction on Algorand
- **2nd Runner-up**, IIT Bhubaneswar Pravaah Web Hackathon 2026 – built cosmic-watch-mocha.vercel.app to track asteroid risk using NASA APIs

Technical Skills

System Design: rate limiting, anti-scraping, prompt-injection defense, OTP/OAuth hardening, state machines, RBAC
Agentic AI: multi-agent systems, agent governance, LLMs (Anthropic on AWS Bedrock, Gemini, Sarvam AI)
Languages: JavaScript, TypeScript, Java (DS&A), HTML, CSS
Frontend: React, Next.js, Tailwind CSS, Vite
Backend: Node.js, Express.js, REST APIs, JWT, OAuth, Firebase Auth, Clerk, Razorpay, Resend
Data & Infra: MongoDB, SQL, Upstash Redis, Cloudinary, AWS (Bedrock, Location Service), Vercel, Render, Algorand